

Financial Distress Prediction Models: A Review of Their Usefulness¹

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SUMMARY Throughout the 1980s there has been continued interest in developing financial distress prediction models for both large and small firms. There has, however, been no survey of this literature directed towards assessing the uses and limitations of these models in a management context. The purpose of the paper is, therefore, to indicate the managerial uses and limitations associated with adopting financial distress prediction models. The paper achieves this end by considering in section two the current financial distress prediction techniques and their limitations. Section three examines the relevance of the predicted event (usually actual failure), the usefulness of multi-outcome models and the appropriateness of various sample selection methods. A review of the range and adequacy of the financial and non-financial information used to construct predictive models forms the basis of the fourth section and is followed in section five by a review of the validity of the claims made on behalf of their predictive accuracy. The following section examines the efficacy of other methods of predicting distress, and reviews the 'man versus model' literature concerning the relative abilities of unaided human decision-makers and statistical models. The final section offers conclusions and suggests where future work might be directed.

1. Introduction

Over the past 30 years, a vast literature has emerged concerned with the development of statistical models designed to predict if firms will either fail or experience some other less severe form of financial distress, such as loan default or the non-payment of creditors. On the face of it, these models appear to be highly successful, in that most published large firm studies report that their models are able to correctly predict the outcome of approximately 95% of firms, 1 or more years prior to the criterion event. The literature on large firms has been well surveyed in a number of specialist review articles (see Scott, 1981; Argenti, 1983; Dambolena,

1983; Zavgren, 1983; Taffler 1984; Whittred and Zimmer, 1985; and Jones, 1987) and accounting/finance texts (see Altman *et al.*, 1980; Altman, 1983). In addition, predictive models specifically developed for European countries are described in a special issue of the *Journal of Banking and Finance* (1984). Reviews of the emerging small firm failure prediction literature can be found in Storey *et al.* (1987) and Keasey and Watson (1991). Given that these reviews have tended to concentrate upon the statistical underpinnings and the relative predictive ability of the various models used, the present review does not attempt a comparison of existing models.

Instead, due to the fact that the interest in developing financial distress prediction models seems to be largely motivated by the perception that an early warning of impending financial distress will confer large benefits to a number of parties with

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an interest in the firm, there would seem to be a need to consider the decision usefulness of these models. For example, management, shareholders, lenders and auditors may each be able to take actions that reduce/avoid the costs which would be incurred if the firm failed without 'adequate' warning. In terms of evaluating the decision usefulness of the existing predictive models, the main questions that need to be addressed are: 'Are the statistical models capturing the dimensions of financial health which are important to the decision context?'; 'Do they work better than other techniques?'; 'Do they work consistently over time?'. and 'Can the models be improved upon?' Despite the importance of such considerations, these issues have yet to be explored in any detail. The primary purpose of this review is, therefore, to address these issues and to provide an indication of the main benefits, limitations and potential future developments of financial distress/failure prediction models.

2. Statistical Methods

From the outset it is important to recognize that the statistical models described in this section do not constitute an explanatory theory of failure/distress. Rather they summarize (via statistical aggregation) information contained in a firm's financial statements, to determine whether or not the firm's financial profile most resembles the financial profiles of previously failed (distressed) or non-failed (non-distressed) firms. They can, therefore, be more accurately classified as descriptive tools of a pattern recognition nature. The first attempts to predict financial distress using financial information utilized univariate techniques. With univariate analysis, a financial ratio is computed for the firm of interest, and then compared to a historically derived benchmark for the ratio that is perceived to separate failed from non-failed firms. A basic assumption of ratio analysis is that a proportionate relationship exists between the two variables whose ratio is calculated. The validity of this assumption is crucial if the common practice of comparing a company's ratio to a general standard is to be at all informative. Whittington (1980) points out that two of the conditions necessary for proportionality to hold are likely to be violated in practice. First, there may be a constant term in the relationship between the two variables that make up a ratio.

For example, if the profit/sales ratio is being considered, an element of a company's profit may be unrelated to the sales element, so that the profit: sales ratio may inadequately describe the relationship between profit and sales. Second, the functional form of the relationship may be non-linear. Thus, a company facing decreasing returns to scale, or which was facing a saturated market, might not be expected to achieve a constant increment to profit for each pound sterling added to sales. However, McDonald and Morris (1984) found that, empirically, ratio analysis performed well in capturing the relationships between financial variables. Hence, users may be able to largely ignore possible violations of the proportionality assumptions.

Another more general problem concerns the selection of a reduced sample of ratios. As the classification of companies takes place on a univariate basis, there is potential for conflicting classifications from different ratios. This raises the issue of whether a ratio should be chosen solely on the basis of its predictive accuracy, or whether the consistency of its prediction should also be addressed. Analogous problems also arise in respect of the evaluation of competing multivariate functions (see section 5 on 'Predictive Ability', below).

Approximately 25 years ago, financial analysts began moving away from the univariate use of ratios. The development of multivariate statistical methods able to deal with combinations of two or more variables resulted in the increasing use of multiple discriminant analysis (MDA). In basic terms, discriminant analysis classifies a company into one of two groups (failed/non-failed) on the basis of a statistic (Z-score) that is a weighted combination of ratios that best separates the two groups of firms. The Z-score is derived by assigning weights to the variables, such that the variance between the groups is maximized relative to the within-group variance. To date there are probably in excess of 100 studies that have applied MDA to the prediction of corporate failure. In general these studies have achieved a high degree of classificatory accuracy on derivation and hold-out samples. As will be discussed later, this is not necessarily equivalent to a high degree of predictive accuracy. As the studies are well reviewed in previous surveys, they will not be reviewed here. Instead, some of the potential problems in using the technique will be highlighted.

Discriminant analysis assumes that the ratios are multivariate normally distributed, although

few studies have actually tested for this condition (see Altman *et al.*, 1980 for the statistical underpinnings of the technique). Moreover, linear discriminant analysis (LDA) is theoretically more appropriate when the covariance matrices across the variables are equal for the groups of failed and non-failed firms, whilst quadratic discriminant analysis (QDA) is more appropriate when the covariance matrices for the two groups are unequal. In practice, however, the relative performance of QDA for unequal covariance matrices declines when the sample size is small and the number of explanatory variables is large relative to the sample (see Marks and Dunn, 1974). In fact, LDA has been found to be robust to violations of the assumptions of normality and equal covariance matrices when there is a large distance between the centroids of the categories, and where the distribution of the explanatory variables is similar across the categories. Finally, both LDA and QDA are sensitive in their classification accuracies to differences in covariance matrices, sample size, and the number of explanatory variables.

The wide use of MDA suggests that it possesses a number of clear advantages to either the model builder or the potential user. The ready availability of computer packages incorporating the technique is clearly an important factor in explaining its popularity, particularly since most packages provide a measure of a function's relative performance via its classification accuracy. Even so, MDA has some glaring weaknesses. For example, unless the user is concerned solely with prediction, a major disadvantage is that MDA does not allow the significance of individual variables to be determined. An absolute test of the significance of individual variables is not practical, because the coefficients in a discriminant analysis can take on any value provided the ratios between the coefficients of the variables are maintained. Another assumption that needs to be made when evaluating a MDA model, or for that matter any statistical failure prediction technique, concerns the relative costs associated with the two types of misclassification error. Most studies, by evaluating models solely in terms of their overall predictive accuracy, have assumed equal costs for both misclassified failed and non-failed firms. For economically efficient decision making, however, the predictive functions should be customized to reflect the relative costs to the decision maker of the two types of misclassification. For instance, Anderson (1958) derived an

assignment of observation to group function that directly allowed the costs of the two types of misclassification to differ. Despite the development of general procedures for incorporating costs of misclassification into the classification process, few papers have included differential costs of misclassification (notable exceptions are the papers by Altman 1977 and Altman *et al.*, 1977).

Several more recent studies (for large firms see Martin, 1977; Ohlson, 1980; Mensah 1983; Zavgren, 1985, 1988; and for small firms see Keasey and Watson, 1987b; Peel and Peel, 1987; and Storey *et al.*, 1987) have used probit and logit techniques which provide a conditional probability of an observation belonging to a category, given the values for the independent variables for the observation in question. Both of these techniques use cumulative probability distributions and do not require the independent variables to be multivariate normal. Like MDA these techniques weight the independent variables to obtain a score for a given observation. Where they differ from MDA is that the weights are applied so as to maximize the joint probability of failure for the known failed firms and the probability of non-failure for the healthy firms; they also have the ability to determine the significance of individual variables and they do not have the same demanding assumptions as MDA. However, as there appears to be little difference between the various techniques in terms of model accuracy (see Hamer, 1983), such statistical considerations are likely to be of minor importance for those users concerned solely with predictive power.

An interesting application of the logit technique, which raises the thorny problem relating to the dating of the criterion event (see 'The Choice of Prediction Variables', section 4 below), was made by Zavgren (1988). Taking the probabilities of failure for the various years prior to failure, she developed an entropy measure of the information contained in these signals of impending failure. She found that, as failure approached, the information provided by the logit functions increased. However, Zavgren based this conclusion on results that were derived from the assumption that a given year prior function was correctly matched to the relevant year of prior data. In practice, of course, decision-makers would not have the benefit of hindsight to achieve such matching – if they had, then a predictive model would be redundant. Keasey and McGuinness (1990) replicated the Zavgren study but adopted rules of thumb for matching the data

and functions. Perhaps not surprisingly, they did not find information to increase as failure approached.

Another interesting extension of the logit technique has been provided by Peel and Peel's (1988) use of the multilogit technique. The technique is essentially the same as bivariate logit, except that probabilities can be defined for more than two categories of interest (see Pindyck and Rubinfeld, 1981, for a detailed exposition). Peel and Peel use this unordered multi-category technique to:

'... discriminate simultaneously between healthy firms and failing firms, one, two and three reporting periods prior to failure. Hence, the output from the multilogit model will simultaneously generate the probabilities of a company remaining healthy, or failing one, two and three reporting periods into the future' (p. 310).

The development of multilogit functions for dating the failure of firms assumes that the data available from the various years prior to failure give consistent signals. When this issue was empirically examined by Keasey, McGuinness and Short (1990), they found, however, that the various years of data did not provide consistent signals for the failing firms.

In summary, the various statistical techniques are able only to optimally weigh the information provided. The appropriateness of the resulting predictive functions will be crucially dependent upon the assumptions made regarding the costs of misclassification and the structure and availability of the data. However, there is little doubt that logit/probit analysis offers as much as any other technique to the user. In addition, because of its regression format, it offers ease of interpretation and is readily available via a number of standard packages. There seems little to be gained, given the results of existing empirical work, by the user searching out the more exotic statistical techniques (e.g. polychotomous probit, survival analysis or classification trees).

3. Model Construction and Data Limitations

In order to construct a model, the researcher has to make a series of decisions relating to the criterion event and the sample of firms to be used to calibrate the model. These decisions will often have important implications for the user of the model, in that

they will affect the range of events and class of firms to which the model is applicable.

The Criterion Event

In deciding whether to use a predictive model, the potential user needs to consider if the prediction event of the model is appropriate for the decision context. The majority of predictive models have defined the criterion event to be creditors' compulsory and/or voluntary liquidation (or, in the US, bankruptcy). This is because these definitions of corporate failure, particularly when significant employment and financial losses are involved, are highly visible legal events that can, therefore, be objectively dated. Nevertheless, there are a number of problems with using this criterion.

Whilst there may be parties directly interested in failure, it is also likely that for many decisions the primary interest will be in the forms of financial distress that precede actual failure. Clearly, different forms of financial distress, for example the non-payment of creditors or a bond default, will have varying degrees of correlation with actual failure. Hence, different interested parties may perceive generally defined failure prediction models as having varying usefulness. The potential user of these models needs to ask, therefore, if the criterion event is of direct relevance to his/her decision context, or whether it is possible to develop models with more direct applicability. For example, does the bank interested in predicting loan defaults rely on a failure prediction model, or does it develop a loan default model?

There is some evidence that firms experiencing less severe forms of financial distress can be distinguished from failed firms. Lau (1987), for example, developed a multilogit, 5-state, financial distress prediction model that was able to successfully classify the different categories of distress for his sample of firms. The criteria for determining the financial state category each firm fell into, from least to most distressed, were: the omission or reduction of a dividend; loan default; protection under Chapter X or XI of the US bankruptcy code; and formal liquidation. All firms not falling into any of the above categories were classed as financially stable. Lau's study suggests that event-specific models may be of more relevance to users than a general distress/failure model. The necessarily incomplete and arbitrary nature of any criterion to classify states of financial distress and the

assumptions concerning the relative severity of each financial distress state are, however, important methodological issues that still need to be addressed.

Moreover, reliance upon legal criteria highlights the importance of the institutional and regulatory framework within which firms operate, and the fact that legal regulations are subject to changes which radically alter the type, incidence and costs associated with particular forms of financial distress. For example, in the UK the 1980s was a decade that experienced a number of important changes, such as financial deregulation, an upsurge in corporate restructuring activities (i.e. buy-outs, privatizations, mergers, takeovers, etc.) and the passing of the 1986 Insolvency and Disqualification of Directors Acts. These changes have created a number of new offences (e.g. wrongful trading), financial instruments (mezzanine, strip financing, etc.), new financial markets (e.g. the Unlisted Securities Market), higher gearing ratios (i.e. leveraged buyouts) and a new legal means by which distressed firms can obtain protection from creditors (administration). Given the extensive nature of these institutional changes, models based upon data from earlier time periods will have to be used with care.

Finally, it should be noted that a predictive model is only suitable for those firms which exhibit symptoms of financial distress for some considerable period prior to occurrence of the criterion event. Hence, such firms are to a large extent a self-selected group because, prior to the event, it is always possible in principle for the firm to seek accommodation with its creditors and thereby avoid further dissipation of firm value and/or any deadweight bankruptcy costs. For instance, when a firm begins to experience financial difficulties and a real possibility exists that it will default on, say a bank loan, a refinancing package, a restructuring of its assets, a change in the scale or scope of its operations, or a merger with another firm are all possible alternatives to actual default. A number of recent studies have examined the possibility of a merger/bankruptcy alternative for financially distressed firms (see Shrieves and Stevens 1979; Pastena and Ruland, 1986; Peel and Wilson, 1989; and Peel, 1990. The studies by Shrieves and Stevens, and Peel and Wilson indicate that a significantly larger minority of merged firms, around 15–17 per cent, exhibited symptoms of financial distress in the year prior to merger than did the general population of firms, where less than 5 per cent were

defined as distressed. Moreover, the evidence suggests (see Peel, 1990, for a review) that it is usually the distressed firm which actively seeks a partner. The main *a priori* arguments put forward as to why it is often in both firms' interests to merge concern the value gains to both firms' shareholders, although not necessarily the distressed firm's managers (see Pastena and Ruland, 1986), arising from the increased debt-carrying capacity of the merged entity and the avoidance of the deadweight bankruptcy costs associated with legal liquidation.

Given these considerations, it is difficult to avoid the conclusion that there may be a need to develop specific models for different types of financial distress. This would focus attention on the direct issues of interest and the process that leads the stakeholder (or coalition of stakeholders) to 'pull the plug' rather than instituting some alternative solution, such as the restructuring or refinancing of the firm. This would, of course, require a more explicit theory of failure processes than has typically been the case in these studies (see 'The Choice of Predictor Variables', section 4 below).

Sample Derivation

For most decision contexts the potential user of a financial distress prediction model will have a given population of firms in mind. For instance, in assessing credit worthiness the relevant class of firms would be those with similar characteristics to current credit customers. Before using any failure prediction model, the user needs to establish how good a match the sample underpinning the derivation of the model is to the population of interest. While the final answer to this question must always be empirical, on *a priori* grounds there is often room for doubt. For example, it is unlikely that a model developed for small or medium sized firms will have much predictive worth for large firms. Bearing this in mind, the methods that have been used for deriving samples of failed and non-failed firms are now described.

The large firm failure literature has generally used delistings from publicly available data sets, such as Moody's Industrial Manual and the Compustat Industrial Files (in the US) or Datastream and Extel (in the UK) to derive the sample of failed/distressed firms. As these data sets cover the range of large firms, the user of the predictive model only needs to ensure that the sample used to derive the model has no major industry biases because of its

time frame. As well as ensuring that the prediction model has no industry/time-specific sampling biases, the user needs to ensure that derivation sampling has applicability to his/her decision context.

If the user is interested in the small firm sector, then deriving a sample of failed firms is often a more difficult task because the publicly available data sets are heavily biased towards large firms. For example, the work of Peel and Peel (1987) on private company failure utilized the Extel Unquoted Companies Service to derive their sample of failed firms. This data set is, however, comprised of the 2100 largest private companies in the UK by reference to turnover, which can hardly be said to be representative of small firms in general. If a more representative small firm sample is to be analysed, then the researcher has to create his/her own data set from local sources or client records, or use publications such as the *London Gazette* if the research is UK based.

Once a sample of failed firms is obtained, a control sample of either non-failed or non-distressed firms must be formed. It has been argued (Taffler, 1982) that the control sample should include only non-distressed firms, because currently non-failed but distressed firms will have similar characteristics to the failed sample. Whilst this is likely to increase the predictive power of the resulting model, the absence of clear criteria to distinguish distressed from non-distressed firms creates additional problems. Indeed, the user may require a statistical model to discriminate between non-failed healthy and non-failed distressed firms. This was the strategy adopted by Shrieves and Stevens (1979), who used Altman's Zeta model to identify those merged firms that were financially distressed.

Ideally, the control sample should be a random sample of non-failed (or non-distressed) firms with data covering the same year(s) as the failed sample. This rarely happens for a number of reasons. First, it is unclear what size of sample the random selection should have. Second, a random selection would be likely to result in the non-failed sample containing different proportions of firms from particular industries and size bands than the failed sample. Hence, differences in the values of the independent variables between the samples could not be solely attributed to failure/non-failure. Most studies have dealt with this problem by matching the non-failed firms to the failed firms in terms of industry and size. Such an approach also overcomes the issue of how to define the size of the non-failed

sample. However, this solution is not without its problems, since it immediately rules out size and industry as predictor variables. Furthermore, the use of 'matched samples' to derive the original model (and to ascertain its predictive ability) for populations of firms where the percentage failing is low, may give rise to seriously misleading indications of both a model's 'external validity' and its likely practical value for decision-making purposes. Palepu (1986) suggests that the use of non-random samples has three drawbacks that make the reported predictive results unreliable:

'First, the use of non-random samples in the model estimation stage, without appropriate modifications to the estimators, leads to inconsistent and biased estimates. This results in overstating the model's ability to predict. Second, the use of non-random samples in prediction tests leads to error rate estimates that fail to represent the model's performance in the population. Third, the use of arbitrary cut-off probabilities in prediction tests makes the computed error rates difficult to interpret' (p. 31).

Nevertheless, in an empirical context Zmijewski (1984) found that, while non-random samples gave rise to biases, the biases did not appear to materially affect the overall classification rates. In fact the literature on choice-based sampling (see Manski and McFadden, 1981) suggests that non-random sampling at the estimation stage can have positive benefits. A choice-based sample can often provide more precise estimates than random sampling for a given sample size. Whilst the samples may be of equal size, they can be matched or not matched on other dimensions of interest. Finally, choice-based sampling and estimation procedures have the potential to avoid the problem of overfitting the predictive function to the derivation sample as compared to the population as a whole. With overfitting, there is no guarantee that the model, though faithfully reflecting the peculiarities of the particular sample from which it was derived, will have any predictive content for the rest of the population. If the set of ratios does not have the same interrelationships for the population of firms for which predictions are needed, the model's predictive ability is likely to be considerably less than its performance in correctly classifying the derivation sample would suggest.

A final part of the usual sample derivation process is to ensure that the non-failed firms have the same annual years of data as the failed firms. The

process usually adopted for achieving this is to 'match' the accounts for the two sets of firms in terms of the periods to which they refer. However, failing firms are likely to delay the submission of their accounts (Schwartz and Menon, 1985), and this is a particularly acute problem with small failing firms, since it is not uncommon for the accounts relating to the 2–3 years prior to failure never to be produced or to be unavailable until after failure occurs (Keasey and Watson, 1988). Therefore, users need to ensure that the claimed predictive accuracies are not based on data that would only be available after the occurrence of the criterion event.

4. The Choice of Predictor Variables

There is little doubt, given the complexities surrounding the interaction of institutional changes, ownership interests and the alternatives to failure, that some understanding of the corporate failure process may be required before the user can make an informed choice regarding the most appropriate model (if any) to use. In terms of theory, most empirical models seem to rely (implicitly at least) upon some variant of Beaver's (1966) cash-flow model, which views the firm as a pool of liquid assets which is drained and fed by the firm's operations. Failure occurs when the reservoir runs dry, i.e. firms fail when they run out of money. However, as Bulow and Shoven (1978) have shown, it is not necessarily true that insolvent firms invariably fail and solvent firms survive. Much depends upon the relative claims, economic interests and power of the different stakeholders and their ability to form a dominant coalition capable of managing the situation to further these interests. Hence, these cash-flow models assume a simple mechanical relationship to exist between a firm's financial condition, as reflected in its financial ratios, and the criterion event. There is a clear need, therefore, to understand and disentangle the following two issues:

- (a) the economic process(es) whereby firms become insolvent; and
- (b) how the agents (bankers, creditors, etc.) whose actions determine the firm's fate, actually decide that a firm's financial condition and prospects are insufficient to justify continued support.

However, due to the lack of a formal theoretical model of the relationships between the failure process,

financial variables, and the economic interests and actions of agents, empirical model builders have, to date, generally adopted the strategy of computing a large initial set of ratios and then letting statistical methods reduce the set. For example, mechanical forward or backward stepping procedures that add or delete ratios according to changes in F-values, as compared to a pre-determined level, have been used in a wide variety of papers. The main problem with these mechanical approaches is that statistical overfitting often occurs.

In addition to the potential for overfitting to occur, there is the issue that the basic inputs of the models (values of the financial ratios) are, of course, heavily influenced by the (essentially arbitrary) accounting policies of firms. As accounting numbers are, at best, 'noisy signals' of underlying 'economic realities', several studies have included variables other than traditional accounting ratios. One obvious extension, given the existence of cash-flow models, is the inclusion of cash-flow variables. Casey and Bartczak (1984) specifically considered the predictive ability of cash-flows from operations for large firms. They concluded that cash-flow variables did not provide information over and above accrual-based ratios. Again, this result should not be too surprising, since empirically cash-flows and accounting profits are usually very high correlated.

The results of Casey and Bartczak (1984) and Gentry *et al.* (1985) indicate, however, that there could be some benefit from researching the predictive ability of the variability of the cash flows. Whilst there have been no studies on the information content of the variability of cash flows, some analysis of the variability of traditional accounting ratios has been undertaken. Dambolena and Khoury (1980) analysed a matched sample of 46 failed and non-failed companies taken from Moody's Industrial Manual. They evaluated the effect of including ratio stability measures such as the standard deviation, the standard error of the mean and the coefficient of variation within a discriminant function. Dambolena and Khoury's profiles of stability measures showed striking differences between failed and non-failed companies. Furthermore, they concluded that the inclusion of such measures in a discriminant function substantially improved the results. Similar results have been achieved by Betts and Belhoul (1987) for a sample of firms in the UK. However, such conclusions must be interpreted with care, because they are

based upon a comparison of discriminant functions that include accounting ratios for a single year with functions that include stability measures, which by definition incorporate information from more than one year. The improvement in predictive power ascribed to the stability measures could, therefore, be purely a function of including more years of information rather than the stability measures *per se*.

As firm performance and rates of failure are often correlated with levels of general economic activity, another obvious extension is to include variables that reflect macroeconomic conditions. Such work has been conducted on large firms by Altman (1983) and Cuthbertson and Hudson (1990). Whilst these studies differ in terms of which variables have predictive content, taken as a whole they indicate that the prior probability of failure used in the derivation of the predictive functions could be better conditioned if the health of the economy was allowed for. However, for this to be useful in a distress prediction context, accurate forecasts of macroeconomic variables will need to be generated – a notoriously difficult undertaking if the forecast horizon is more than a few months. Moreover, if macroeconomic variables are indeed important predictors of failure, then the assumed temporal stability, and hence the predictive usefulness, of current models must be thrown into some doubt.

Due to the widely differing impact that prices changes have upon various accounting variables, a number of studies have adjusted historic cost accounting ratios for either general price level changes (inflation) or for specific price changes (current cost accounting). It has been argued that current value accounting may be of special relevance for failure prediction purposes, because it provides a closer proxy for the opportunity costs associated with the current utilization of assets (Keasey and Watson, 1986). Norton and Smith (1979) found that general price level-adjusted ratios were neither more or less accurate than historic cost ratios for the prediction of large firm failure. In contrast, Ketz (1978) found price level-adjusted ratios to be useful when the relative costs of the two types of misclassification were considered. In a more recent study, Mensah (1983) adjusted financial statement data using asset specific price indices. The general conclusion drawn by Mensah was that specific price-adjusted ratios added incremental information over and above that provided by historic cost ratios. For small firms, however, Keasey

and Watson (1986) found that the current cost information did not improve the predictive accuracy of the models. They showed that this was due to the fact that current cost adjustments did little to alter the variance within and between the distributions of failed and non-failed firms. Indeed, because of the high correlation between them, the historic cost and current cost data sets were almost perfect substitutes, and that the current cost adjustments merely ‘rescaled’ the historic costs ratios. Other studies of inflation accounting have come to similar conclusions (see Thomson, 1984, and Thomson and Watson, 1989).

Given the large body of empirical evidence relating to the informational efficiency of the capital markets, it is not surprising that some writers have suggested that incorporating market data into failure prediction models is likely to provide substantial benefits, particularly in respect of timeliness (see Foster, 1986, for a review). This is because the efficient capital markets literature suggests that financial market variables efficiently incorporate expectations and should, therefore, be good ‘lead indicators’ of future events. A number of authors (Altman *et al.*, 1977; Marais *et al.*, 1984; Zavgren, 1988) have examined this proposition in respect of failure prediction. Marais *et al.* found that six non-accounting variables (including share price movements, bond ratings, commercial paper rates, etc) contained as much information as traditional accounting ratios. Zavgren found that failures were not unanticipated, given market data. The large body of empirical evidence relating to the informational efficiency of the stock market implies that incorporating market data into failure prediction models is likely to provide substantial benefits, particularly with respect to timeliness of information.

A number of studies (Keasey and Watson, 1987b, 1988; Peel and Peel, 1987) have examined the predictive content of qualitative variables, such as the lag in submitting accounts, for small firm/private company failure models, and found them to have significant incremental information content. Whittred and Zimmer (1984) carried out similar tests on large Australian firms but found no incremental information content, primarily because of the high level of predictive accuracy of the financial ratio model. For small firms, however, the predictive content of financial ratios is typically considerably less, which gives submission lag and other ‘qualitative’ variables more scope to make a contribution.

More comprehensive studies of the information content of 'qualitative', i.e. non-financial, ratio information, have been undertaken by Peel, Peel and Pope (1986) for large firms and Keasey and Watson (1987b) for small firms. The Peel, Peel and Pope study found a number of non-financial variables to be significant predictors of failure, though no attempt was made to explicitly link these variables to a theory of firm failure. The Keasey and Watson study, however, extracted a number of empirically testable hypotheses from Argenti's (1976) model of failure relating to the following four aspects of small companies:

- (a) the management structure;
- (b) the inadequacy of the accounting information system;
- (c) the manipulation of published financial statements; and
- (d) gearing.

Argenti argues that financial ratios, because they are merely 'symptoms' of business failure, are unable to yield significant insights into the underlying processes or 'causes' of corporate collapse. Argenti recognizes that financial ratios are open to various forms of manipulation and are therefore likely to become less reliable as failure approaches. Managers of failing companies are motivated to engage in all manner of 'creative accounting' practices in an attempt to hide the company's poor financial and trading condition from outside investors and creditors. Argenti views the failure process as being driven by a number of inherent defects in the actual organization and financial structure of the company. For instance, due to the lack of access to equity capital funds, many small businesses are largely financed by short-term bank loans and overdrafts. This form of financing is inherently risky – leaving such firms highly vulnerable to downward changes in economic activity, or, as Argenti puts it: 'high gearing and an economic downturn are the classic nutcrackers of failure' (p. 136).

Other inherent defects include the lack of an adequate product and financial resource base. The major inherent organizational and management defect identified by Argenti is the 'autocrat' or 'one-man band', where a single individual dominates the board of directors and rarely heeds the advice of others working within the enterprise. Thus, Argenti's model of failure is based upon an understanding of the business and management struc-

tures of a firm. The long-term prediction of failure has to allow for these fundamental factors, for they, by definition, precede the consequent financial symptoms.

The results of empirically testing Argenti's theory of failure on small firms (Keasey and Watson, 1987b), to which it seems most suited, indicated that models containing the non-financial ratio information were more 'robust' and significantly out-performed the model utilizing financial ratios alone. The predictive worth of qualitative factors has also been considered by von Stein and Ziegler (1984) for a sample of medium-sized firms in Germany. In a highly innovative paper, von Stein and Ziegler considered bank data and 'management' characteristics as potential predictors of distress. The 'management' characteristics considered included such factors as worker dismissals, changes in legal form, decreases in demand, competitor pressure, production details and the personal profiles of the managers. As well as indicating the importance of understanding business structure for failure prediction, these 'qualitative' models appear to have certain practical advantages. The non-financial variables, unlike many of the financial ratios, are less likely to suffer from the vagaries of manipulation and off-balance-sheet financing activities, and items such as notification of changes in directors and secured loans are more timely than annual accounts.

In terms of large firm failure prediction, the available evidence suggests that predictive accuracy is reasonably insensitive to the financial ratios chosen. The empirical evidence for small-and medium-sized firms indicates, however, that there are benefits, in terms of predictive accuracy, from understanding the underlying business structures that create crises out of normal business hazards. It seems likely, therefore, that a more in-depth examination of the 'process of failure' and of the decision processes of the actors involved may lead to more informed and longer-term predictive success.

5. Predictive Ability

The majority of the studies have measured the success of their predictive functions in terms of their classificatory accuracy on hold-out samples. These external tests of a model's validity are vitally important to determine whether it is a genuinely predictive model or merely an autopsy of pre-

viously deceased firms. A number of studies have used cross-sectional hold-out samples, that is, the test sample is from the same time period as the derivation sample. Of course, such a procedure will produce an indication of the predictive accuracy of a model only if the determinants of failure are stationary over time. A preferred test procedure is clearly one where a future dated hold-out sample is used, since this would provide a direct test of the temporal stability of the model. When Altman and McGough (1974) conducted a future dated validation of Altman's 1968 discriminant model, using a sample of 34 bankrupt firms from the 1970–1973 period, the findings indicated that 82.4 per cent of the observations had been correctly identified from one year prior data. A similar 'the proof of the pudding is in the eating' test is reported in Taffler (1984). Taffler notes that his 1976 model indicated that 115 of the 825 firms on the Exstat data base post 1976 were defined as being at risk. Of these 115 firms identified as being at risk, 50 were found to have suffered financial distress in the 6 years following model development. A problem with both of these tests is their sole concentration on the firms that were defined to be at risk. When Moyer (1977) applied Altman's 1968 model to 1 year's prior data for both failed and non-failed firms for the 1965–1975 period, he found the model to have only modest predictive accuracy (75 per cent predictive success as compared to Altman's original 96 per cent). It is not possible to argue that this result solely reflects the instability of Altman's predictive function, because the firms sampled by Moyer were larger than those used by Altman for the derivation of the model. Nonetheless, work by Mensah (1984) suggests that bankruptcy prediction models are fundamentally unstable, in that the coefficients of a model will vary according to the underlying health of the economy. These various results suggest that model derivation should be as close in time as possible to the period over which predictions are to be made.

Moreover, for any hold-out sample results to be representative of the kind of success a decision-maker could achieve in practice, the sample must represent the population the decision-maker will face when having to reach a decision. If the population of firms that the decision-maker is interested in is characterized by very few failures, then a matched sample design is likely to overestimate the success rate that will be achieved in practice (see Palepu, 1986):

A further shortcoming of most studies has been the lack of attention given to either the timing of the criterion event or the information a user would have available at the time a decision is required. Certain studies have developed different functions for the various years prior to failure and then tested their predictive accuracy by correctly matching function to data for a hold-out sample. Clearly, in a real prediction situation no such matching could occur. Moreover, even if only one function is developed, the decision-maker will still normally have more than 1 year of prior data available from which to make predictions. Apart from a study by Keasey, McGuinness and Short (1990), there has been a lack of attention given to the possibility that the various years of prior data may give conflicting signals, even where a single predictive function has been used. In the presence of conflicting signals (and the study by Keasey *et al.* suggests that conflicting signals are not a rare occurrence), the existing literature offers little in the way of advice to the decision-maker. Given the potential for this problem to occur, more emphasis needs to be given to measuring the consistency of model predictions and the dating of the criterion event.

The majority of studies on failure prediction have defined predictive success in terms of a function's ability to correctly classify a sample – where classification success is defined as the joint minimization of type 1 and type 2 errors. This average (across the two types of error) classification success may not measure, however, the usefulness of a predictive function, because the decision maker may be more concerned with correctly predicting failure than non-failure and/or non-distress. Part of this problem stems from the lack of specification of the decision context. Existing models appear to have been developed with some abstract 'general' decision-maker in mind. However, without more specified decision contexts it is difficult to judge the actual usefulness of these predictions.

Furthermore, as it is difficult to ascertain how successful the decision makers are with their present information and methods, the extent of the potential benefit of the existing models is also unclear. Since there is considerable ambiguity regarding the information available and used by the various decision-makers, it is difficult to determine if the information is being used optimally or if there is additional information available that could be usefully applied to the decision. The lack of specification of the decision context also makes

it difficult to assess whether the decision-maker in question would be able and willing to adopt a new technique, given his/her investment in existing technology. To date, the literature has concentrated on an abstract/general decision context and, if the aim is to materially improve practical decision-making, which presumably is one of the main motivations behind this research, then greater attention needs to be directed towards the above issues.

6. Alternatives to Statistical Distress Prediction Models

The perceived benefits from using statistical prediction models will be heavily influenced by the efficacy of existing methods of predicting the event of interest. Basically, if these existing methods, which may consist of only an unaided human decision-maker, are poor at predicting failure, then the expected benefits from using a model with a given level of accuracy will be much greater than when existing methods are more efficacious. There is a large literature (see Libby and Lewis, 1982; Houghton, 1984; Whittred and Zimmer, 1985, for surveys) that has investigated the extent to which various user groups, such as credit raters, loan officers and auditors, can accurately predict failure from a set of financial ratios. An early and highly influential piece of work in this literature was provided by Libby (1975). Libby's results indicated that the loan officers were able to distinguish between failed and non-failed firms with an accuracy of 74 per cent. Other studies on large firms have considered a variety of agents (for example, Zimmer, 1980; Casey 1983; Kida, 1980) and have achieved success rates at least equal to Libby's – although subjects using small firm data have not been able to achieve such high predictive success rates because of the poor diagnostic content of the financial information supplied by small firms (see Keasey and Watson, 1987a). This literature has also shown, however, that a statistical model is usually able to significantly outperform specialists (see Dawes and Corrigan, 1974; Houghton and Sen-gupta, 1984). This ability of statistical functions to outperform human decision-makers (given the same information set) merely reflects the fact that statistical models are able to optimally weight the individual components of an information set and are more reliable, that is, they are more consistent, than even the most 'expert' of human information

processors.

However, the above research findings may not be an appropriate benchmark by which to judge the relative predictive capabilities of human decision makers and statistical models (see Einhorn and Hogarth 1981). Generally, studies of the above type have provided their subjects with highly constrained information sets – typically a number of financial ratios – much less than they would have available in a real decision situation. Moreover, much of the additional information typically used by human decision-makers is of a non-quantitative nature which is often unsuitable for incorporation in the types of statistical models examined in this paper. One area where human decision-makers may have an advantage over statistical models is in the specification of the prior probabilities of failure. Their detailed knowledge of the market place and the firms of interest may lead to reasonably precise prior probabilities of failure. Whilst subjective prior probabilities of failure are, of course, potentially quantifiable, the very act of quantification may distort their integrity.

Given the above, a more informative evaluation of human decision-makers would be to allow human subjects to make predictions after having access to all the information they usually use when coming to a decision. A further criticism of these studies, which purport to show that models outperform humans, is that they assume that decisions are made by isolated individuals. In fact, many of these decisions are made by interacting groups, committees, etc. Hence, the relevant benchmark will not be the predictive accuracy of an isolated individual's decisions, but rather the aggregated judgments or consensus forecasts of interacting groups, which, as Solomon (1982) and Chalos (1985) have shown, are generally significantly superior to both individual predictions and those of linear models. Indeed, Chalos concluded that committees had:

'significantly fewer and less costly loan action errors than the individuals. Moreover, the marginal benefit of committee processing accuracy clearly appeared to outweigh the incremental cost of committee time' (p. 540).

7. Concluding Remarks

This paper has reviewed the existing literature on financial distress prediction with the main emphasis

being placed upon the difficulties associated with assessing their usefulness. The purpose of this conclusion is to evaluate the current position and to suggest directions future research might take. There seems little doubt that additional studies utilizing existing methodologies are likely to produce increasingly poor returns in terms of improving either our understanding of firm failure or in achieving significantly superior predictive accuracy. There is little to be gained from further studies that merely replicate the existing general methodology with additional 'tweaks' in terms of data and/or statistical method. It is clear that reasonable degrees of 'classificatory accuracy' (see Hamer, 1983) can be achieved from a wide range of ratios and statistical techniques. If the literature on distress/failure prediction is to progress further, then more explicit and formal modelling of the economic interests and decision processes of the firm's major stakeholders will probably have to be undertaken.

The main context where the present general models may provide material benefits seems to be where a relatively cheap and simple-to-use preliminary screening device for routine credit/lending decisions is required. This is because, for many well-specified and repetitive decisions, the classification accuracy of even relatively simple quantitative models have been shown to consistently outperform individual human decision-makers (see Houghton, 1984, and Libby and Lewis, 1982, for a review of the evidence). Moreover, once the initial costs of constructing the model have been incurred the incremental costs associated with its continued use will be relatively small – although, if temporal instability proves to be a problem, additional costs will be incurred whenever the model is recalibrated. Hence, the benefits obtained from using a model do not have to be very large for it to be a worthwhile, that is, cost-effective, tool. If a predictive model is, however, required as an input into more strategic decision-making, then the existing empirical models are likely to continue to play a very minor role. Moreover, the evidence (see Ray and Hutchinson, 1983, Storey *et al.*, 1987; Hutchinson, Meric and Meric, 1988) suggests that existing predictive models are unable to distinguish between highly successful but financially strained firms and those in serious financial distress – although models developed along the lines of the multi-outcome and merger/liquidation alternatives may eventually form a greater input.

Future work on failure/distress prediction ought

therefore to focus more directly upon the needs of the decision-maker. There is a need to understand how he/she presently reaches a decision for the context in question. This entails defining the decision to be made, the information currently used and the method adopted for reaching a decision. Once this is thoroughly understood, more useful decision aids may be forthcoming. Given the absence of an acceptable theory of failure/distress (however defined), a thorough understanding of the present processes used by 'experts' should inform model development. The existing literature on modelling human decision processes indicates that the major benefit of statistical techniques over individual experts is their reliability. Human experts, however, are able to access a wider range and variety of information inputs and to process it in a manner more suited to the specific decision context in question. They are also able to call upon the knowledge of other experts and to reach better decisions working collectively. A more thorough understanding of the relative strengths and weaknesses of humans and statistical models in a wide variety of decision contexts is required if the next phase of the failure/distress prediction research programme is to make significant improvements to existing models.

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